MILITARY SPECIFICATION
TEST STATION, COMPUTER
OJ-216( )/AWM-23(V)

This specification has been approved by the Naval Air Systems Command, Department of the Navy

1. SCOPE

1.1 Scope. This specification covers the performance, design, and test requirements for the Computer Test Station (CTS) OJ-216( )/AWM-23(V) (see 6.1).

1.2 Classification. The CTS consists of a 4-bay station as follows:

a. Bay No. 1: Console, Test, Computer OJ-256( )/AWM-23(V)

	Nomenclature	Reference Paragraph
(1)	Dummy Load, Electrical DA-633( )/AWM-23(V)	3.6.1.1
(2)	Panel, Test, Electrical C-9180 ( )/AWM-23(V)	3.6.1.2
(3)	Mounting MT-4504( )/AWM-23(V)	3.6.1.3
(4)	Simulator, Missile Auxiliaries	3.6.1.4
(5)	Power Supply Assembly PP-6851( )/AWM-23(V)	3.6.1.5
(6)	Simulator, Electronic Command Signal Generator SM-657( )/AWM-23(V)	3.6.1.6
(7)	Housing Connector Assembly - Loop Test, Relay Matrix	3.6.1.7

b. Bay No. 2: Console, Test, Computer OJ-257( )/AWM-23(V)

(1)	AC and DC Radiometer - Voltmeter	3.6.2.1
(2)	Programmable Counter/Timer	3.6.2.2

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to:

Engineering Specifications and Standards Department (Code 93) Naval Air Engineering Center, Lakehurst, NJ 08733, by using self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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Nomenclature	Reference Paragraph
(3) Indicator, Data Display IP-1129( )/AWM-23(V)	3.6.2.3
(4) Mounting MT-4505( )/AWM-23(V)	3.6.2.4
(5) Simulator, Data Processor	3.6.2.5
(6) Adapter, Power Supply and Bulk Store-Loop Test	3.6,2.6
(7) Switching Unit, Fault Isolation SA-1898( )/AWM-23(V)	3.6.2.7
(8) Simulator, Electronic Command Signal Generator SM-659( )/AWM-23(V)	3.6.2.8
c. Bay No. 3: Console, Test, Computer OJ-258( )/AWM-23(V)	
(1) Oscilloscope	3.6.3.1
(2) Panel, Test, Electrical C-9181( )/AWM-23(V)	3.6.3.2
(3) Mounting MT-4506( )/AWM-23(V)	3.6.3.3
(4) Computer, Digital CP-1159( )/AWG-9	3.6.3.3.1
(5) Mounting MT-4507( )/AWM-23(V)	3.6.3.4
(6) Simulator - Memory Unit	3.6.3.5
(7) Signal Driver Memory Unit	3.6.3.6
(8) Power Supply Assembly PP-6852( )/AWM-23(V)	3.6.3.7
(9) Power Supply Assembly PP-6853( )/AWM-23(V)	3.6.3.7
(10) Power Supply PP-6854( )/AWM-23(V)	3.6.3.7
d. Bay No. 4: Console, Test, Computer OJ-259( )/AWM-23(V)	
(1) Panel, Power Distribution SB-3697( )/AWM-23(V)	3.6.4.1
(2) Magnetic Tape Reader	3.6.4.2
(3) Control, Logic-Magnetic Tape Reader C-9182( )/AWM-23(V)	3.6.4.3
(4) Work Surface Assembly	3.6.4.4
(5) Drawer, Storage	3.6.4.5
(6) Power Supply Assembly PP-6855( )/AWM-23(V)	3.6.4.6
(7) Power Supply Assembly PP-6856( )/AWM-23(V)	3.6.4.6
(8) Power Supply PP-6857( )/AWM-23(V)	3.6.4.6

1.3 Associated equipment. The CTS shall operate with Kit, Tool Special TK-222( )/AWM-23(V).

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2. APPLICABLE DOCUMENTS

2.1 General. The following documents, of the issue specified in AS-4053, Applicable Document List for AWCS Test Set AN/AWM-23(V), form a part of this specification to the extent specified herein. MIL-T-81734(AS) shall be the applicable general specification for the CTS described herein (see 6.6).

SPECIFICATIONS

Military

MIL-M-7793	Meter, Time Totalizing
MIL-T-81734(AS)	Test Set, Airborne Weapon Control System, AN/AWM-23(V)

STANDARDS

Military

MIL-STD-461.	Electromagnetic Interference Characteristics Requirements for Equipment

PUBLICATIONS

Hughes Aircraft Company

3282207	Acceptance Test Procedure, Computer Test Station

2.2 Availability of documents. Applicable (see 6.6).

2.3 Precedence of documents. When the requirements of the contract, this specification, or applicable subsidiary documents are in conflict, the following precedence shall apply:

a. Contract. The contract shall have precedence over any other document.
b. This specification. This specification shall have precedence over all applicable subsidiary documents (see 6.6).
c. Referenced documents. Any document referenced in this specification shall have precedence over all applicable documents referenced therein.

3. REQUIREMENTS

3.1 Materials, parts, and processes. Applicable (see 6.6).

3.2 Design and construction. Applicable except as follows.

3.2.1 Design features. Applicable except as follows.

3.2.1.1 Method of operation. Applicable.

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3.2.1.2 Commercial test equipment. Applicable.

3.2.1.2.1 Other commercial components. Applicable.

3.2.1.3 Circuit grounds. Applicable.

3.2.1.4 Interchangeability. Applicable.

3.2.1.5 Identification of product. Applicable.

3.2.1.6 Human factors. Applicable.

3.2.1.7 Test set generated acoustic noise. Applicable.

3.2.1.8 Forced-liquid cooling. Not applicable.

3.2.1.9 Space and weight limitations. The weight of the CTS shall be less than 4000 pounds. Maximum dimensions shall be 96 inches by 36.25 inches by 75 inches with all drawers and retractable work surfaces in the retracted position.

3.2.1.9.1 Storage space. Storage space shall be provided in two locations in computer bay no. 4. One storage area shall be located to the right of Control, Logic, Magnetic Tape Reader C-9182( )/AWM-23(V) immediately above the work surface, and shall be used to stow any simulator or the slave digital computer weapon replaceable assembly (WRA) when it is removed. The other storage area shall be provided within the drawer located below the retractable work surface. This storage area shall be used for stowage of test leads and magnetic tape reels and will provide some space for manuals. In addition, a minimum of three cubic feet of separate storage facilities will be required to store miscellaneous equipment.

3.2.1.10 Connectors. Applicable.

3.2.1.11 Secondary electrical power. The total secondary electrical power shall be limited to 1900 watts maximum with any combination of power form loading within the load capability limits for each power form listed in Table I. Secondary electrical power shall consist of regulated power forms. Regulation from no load to full load, and for normal values of facility line voltage variations shall not exceed 2 percent at the voltage sense point. Ripple voltage for dc power forms shall not exceed 150 millivolts (mV) peak-to-peak (p-p).

3.2.1.12 Interface. Applicable.

3.2.1.12.1 Clean air interface. The CTS shall interface with the clean air through ducts which enter the CTS at the top rear of each test console.

3.2.1.12.2 Electrical power Interface. The CTS shall interface with electrical power for the convenience outlets at one end of the convenience outlet strip mounted on the top of the front face of the CTS. The CTS shall interface with all other electrical power through power lines entering the top of bay number 4.

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TABLE I

REGULATED SECONDARY POWER FORMS

ITEM	POWER FORM (Vdc)	MAXIMUM LOAD CAPACITY (Amperes)
1	+5	96.7
2	+5.2	26
3	+12	13.4
4	+12.2	3.6
5	+12.5	2.3
6	+15	2
7	+25	14.9
8	+28	2.2
9	+32	9.2
10	+110	0.3
11	-6	5.8
12	-6.2	2.9
13	-12	3.7
14	-15	0.6
15	-20	0.3
16	-25	0.6
17	-32	1.1
18	-110	0.3

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3.2.2 Utilities. Applicable except as follows:

3.2.2.1 Electrical power. Applicable. The CTS shall operate with the electrical power of Table II.

3.2.2.2 Clean air. Applicable. The test station requirement for" forced air shall not exceed 46 pounds per minute.

3.2.2.3 Potable water. Not applicable.

3.2.2.4 Liquid coolant. Not applicable.

3.2.2.5 Helium. Not applicable.

3.2.2.6 Nitrogen. Not applicable.

3.2.2.7 Hydraulic fluid. Not applicable.

3.2.3 Service conditions.

3.2.3.1 Electrical. Applicable.

3.2.3.2 Environmental. Applicable.

3.2.3.2.1 Organizational level. Not applicable.

3.2.3.2.2 Intermediate level. Applicable.

3.2.4. Maintainability. Applicable.

3.2.4.1 Mean time to repair. Applicable.

TABLE II

CTS ELECTRICAL POWER

ITEM	INPUT POWER FORM	MAXIMUM LOAD
1	+29 ± 4 Vdc	0.1 kilowatt (kW)
2	115 Vac, 400 Hertz (Hz), 3-phase, 4-wire, Type III	2.0 kilovolt amperes (kVA)
3	115 Vac, 60 Hz, 1-phase, Type I	1.75 kVA
4	115 Vac, 60 Hz, 3-phase, 3-wire, ungrounded, Type I	4.0 kVA

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3.2.4.2 Test points. Applicable. Test points shall be provided to facilitate fault isolation to the SRA level, alignments, adjustments, and signal monitoring.

3.2.4.3 Shop replaceable assemblies(SRAs). Applicable.

3.2.4.4 Self test capability. The CTS shall contain self test circuits capable of giving the station a confidence check. The self test features shall aid in fault isolation of the CTS. Results of self tests shall be indicated visually to the operator. Duration of the self test shall not exceed 15 minutes. In addition, the CTS shall have the capability to select and perform an appropriate part of the complete CTS self test before testing a WRA.

3.2.5. Reliability. The mean time between failures (MTBF) shall be 134 hours, minimum.

3.2.5.1 Operating life. Applicable.

3.2.5.2 Storage life. Applicable.

3.2.5.3 Operational stability. Applicable.

3.2.6 Electromagnetic interference (EMI). Applicable, with the following exceptions:

a. MIL-STD-461, Tests CE01 and CE03 and the limits therein shall apply.
b. MIL-STD-461, Tests CE02, RS01, RS02, and RS04 shall not apply.
c. MIL-STD-461, Test CE04 shall apply only when power remote control leads external to the test station are used.
d. MIL-STD-461, Test RE04 shall apply except the test shall be limited to the area occupied by the WRAs containing memory assemblies. The magnetic field shall be less than 5 gauss.

3.2.7 Utilities protection. Applicable except as follows.

3.2.7.1 Liquid cooling protection. Not applicable.

3.2.7.2 Clean air protection. Applicable.

3.2.7.3 Electrical power protection. Applicable.

3.2.7.3.1 Secondary voltage protection. Occurrence of an out-of-tolerance secondary voltage condition that could damage either an item under test or the test station shall result in automatic shutdown of the secondary power forms. The out-of-tolerance condition leading to shutdown shall be identified by an appropriate indicator. Operator action shall be required to reinstate test station operation after the occurrence of automatic shutdown.

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3.2.7.3.2 Emergency shutdown switches. Emergency shutdown switches shall be provided for use by the operator. One switch shall be located on the power distribution panel in bay no. 4: the other on the electrical dummy load in bay no. 1. Both switches shall be push-button activated, momentary contact type with the push-button finished in red. The switches shall be superimposed on a bright orange-yellow and black diagonal striped background. Activation of either switch shall result in shutdown of the CTS and the load side of all primary circuit breakers shall be disconnected from their respective loads. Emergency power shutdown shall be fail-safe and Independent of all utility power forms.

3.2.8 Calibration cycle. Applicable.

3.2.9 Power switching and coolant control. The CTS shall provide the necessary switching to control utility power forms and shall provide orifices and flow switches as required for control of clean air.

3.2.10 Displays. Displays shall be provided for:

a. Indicating station status
b. Monitoring the test performance
c. Examining the contents of the computer registers
d. Signaling the operator when operator action is required

3.2.11 Operator controls. Operator controls shall be provided for selecting test options and controlling the automatic test program.

3.2.12 Tape(s). Tape (or tapes) required for the CTS shall be of 1/2 inch Mylar magnetic tape, or equivalent. Maximum reel diameter shall be 8-1/2 inches, CTS maintenance tape(s) shall contain the programs required to verify performance of the WRAs listed in Table III and shall also contain diagnostic test routines for fault isolation in malfunctioning WRAs. CTS maintenance tapes shall also contain programs for self test of the CTS. A different tape containing the tactical and built in test (BIT) programs shall be provided, but not as part of the test station.

3.3 Performance.

3.3.1 General. The CT'S shall provide the capability to test and maintain the equipment listed in Table III.

3.3.2 Organizational level functional requirements. Not applicable.

3.3.3 Intermediate level functional requirements. Applicable.

3.3.4 Warmup time. Applicable.

3.4 Detailed functional requirements.

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TABLE III

WRAs MAINTAINED AT THE CTS

ITEM NUMBER	WRA NOMENCLATURE
1	Computer, Digital CP-1159( )/AWG-9
2	Memory Unit MU-588( )/AWG-9
3	Processor, Data CV-3151( )/AWG-9 and CV-3165 ( )/AWG-9
4	Power Supply-Storer PP-6776( )/AWG-9
5	Converter, Signal Data CV-3152( )/AWG-9
6	Control, Logic-Timing C-9506( )/AWG-9

3.4.1 Maintenance capabilities. The CTS shall be capable of loading program data into the non-destructive read-out (NDRO) of the memory unit and into the bulk store of the power supply - storer memory. The CTS shall provide and control the stimuli and power forms, and provide loads required to perform functional and diagnostic testing of the WRAs listed in Table III.

3.4.2 Test capabilities. Unless otherwise specified herein, functional and diagnostic testing of WRAs shall be performed automatically. The programmable counter/timer and the digital radiometer voltmeter (DRVM) shall be operated under program control (normal) or manual control (optional). The oscilloscope shall be operated only in the manual mode.

3.4.2.1 Computer, Digital CP-1159( )/AWG-9 tests. Operation of the digital computer shall be verified by performance of the following tests.

3.4.2.1.1 Computer self-test. The computer self-test shall verify the performance of the instruction repertoire and shall include:

a. Arithmetic tests
b. Register tests
c. Destructive readout (DRO) memory tests
d. Internal interrupt tests

3.4.2.1.2 Interface tests. The controls and data paths provided for the interfaces listed shall be verified:

a. Data processor interface controlled channel and the program controlled channel.
b. CTS interface

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c. Memory unit interface
d. Power supply-storer (PS/S) interface

3.4.2.1.3 Interrupt and status tests. The response of the computer to external interrupt signals and control shall be verified.

3.4.2.2 Memory Unit MU-588( )/AWG-9 tests. Operation of the memory unit shall be verified by performance of the following tests.

3.4.2.2.1 Address decode. Word and plane address decodes in the memory unit shall be verified by selection of predetermined locations.

3.4.2.2.2 Memory unit control. Word and plane enable tests shall be performed to verify the memory control.

3.4.2.2.3 Memory unit storage. The capability shall be provided to write test patterns into the memory unit, accomplish the preread operation, and read and verify the test data.

3.4.2.2.4 Load and verify tactical program. The capability shall be provided to write the tactical program into the memory and to verify the program by matching all data, bit by bit against the tactical/BIT tape, or if a memory unit write has not been performed, to verify the stored tactical/BIT program by matching all data, bit by bit, against the tactical/BIT tape.

3.4.2.3 Processor, Data CV-3151( )/AWG-9 and CV-3165( )/AWG-9 tests. Operation of the data processor shall be verified by performing the following tests.

3.4.2.3.1 BIT verification. Tests shall be performed to verify proper operation of the data processor self-test circuitry in the WRA under test.

3.4.2.3.2 Interface tests. Each of the following interfaces shall be functionally tested to verify proper operation of the data processor.

a. Program controlled channel - The CTS shall test the data transfers to and from the computer via the program controlled channel.
b. Data processor controlled channel - The CTS shall test the data transfers to and from the computer DRO memory via the data processor controlled channel.
c. Interrupt channel - The CTS shall test the data processor generated interrupt signals. The interrupts to be tested shall include the following operational and tactical interrupts:

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(1) Operational interrupts

(a) Computer test
(b) Program restart
(c) Interval timer

(2) Tactical interrupts

(a) Start sample command (SSC)
(b) Start sample command delayed (SSCD)
(c) Frequency processed
(d) Special standard serial interface (SSI) passed
(e) Special SSI failed
(f) SSI "B" loops completed
(g) Antenna on target
(h) Special analog-digital-analog (ADA) completed
(i) Bulk store end of block

d. Interval timer - The CTS shall test the timing count outputs generated by the interval timer.

3.4.2.3.3 Data processor tests. The CTS shall test the data processor functions. These functions shall include both the communication functions and the special purpose processing functions.

a. Communication functions - Communications functions include the interfaces between the data processor and other avionics units. These functions include the following:

(1) System timer
(2) Radar/infrared (IR) interface
(3) High pulse repetition frequency (HPRF) missile message
(4) Controls and displays interface
(5) Data link interface
(6) Identification friend or foe (IFF) interface
(7) Digital inputs and digital outputs
(8) ADA converter
(9) Standard serial interface (SSI)
(10) Bulk store magnetic tape memory interface

b. Special purpose processing functions - The special purpose processing functions include the interface between the data processor and the radar subsystem. These functions include the following:

(1) Range processor
(2) Acquisition processor
(3) Frequency processor

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3.4.2.4 Power Supply-Storer PP-6776( )/AWG-9 tests. Operation of the PS/S shall be verified by performing tests on both the power supply and bulk store sections of the WRA.

3.4.2.4.1 Power supply tests. The CTS shall perform the following tests to verify that the power supply section of the PS/S is operating correctly:

a. DC voltage outputs
b. Input and output power
c. Power overload
d. +32 Vdc overcurrent
e. Transient detector
f. Ripple and noise

3.4.2.4.2 Bulk storage section tests. The CTS shall provide the capability to perform the following tests to verify that the bulk storage section of the PS/S is operating correctly:

a. Mechanical tape motion verification
b. Magnetic tape memory (MTM) search capability
c. MTM read data capability
d. Write test pattern and verify
e. Load MTM from tactical/BIT tape and verify

3.4.2.5. Converter, Signal Data CV-3152( )/AWG-9. Operation of the signal data converter shall be verified by performing the following tests:

3.4.2.5.1 BIT verification. The CTS shall test the built-in-test circuitry of the signal data converter.

3.4.2.5.2 Reference signal processing. The CTS shall test the 7.1 and 26 Vrms system reference signals.

3.4.2.5.3 Digital-to-analog conversions. The CTS shall test the digital-to-analog (D/A) converters which provide the seeker head positioning data.

3.4.2.5.4 AIM-7 analog transforms. The CTS shall test the AIM-7 analog transform function. This shall include both ac and dc analog output signals.

3.4.2.5.5 AIM-7, AIM-54 parameters. The CT'S shall test the AIM-7 and AIM-54 output parameters generated in the tactical mode of operation.

3.4.2.5.6 Missile message and doppler gravity bias timer. The CTS shall test the missile message and doppler gravity bias timer. This shall include the doppler gravity bias, doppler gravity clock, and the missle address outputs.

3.4.2.5.7 Signal matrix and logic 1 and 2. The CTS shall test the signal matrix and logic 1 and 2 functions. This shall include the mission station selected output and outputs, 10 tactical plus 4 missile on aircraft test (MOAT) signals multiplexed together, which are sent to the missile station selected.

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3.4.2.5.8 Missile-on-aircraft-test (MOAT). The CTS shall test the MOAT logic in the signal data converter. This shall include the missile station selected output (MOAT) and the 7 outputs, multiplexed data from the functional logic-areas in the signal data converter; which are sent to the missile station selected.

3.4.2.5.9 Interface circuit and output logic. The CTS shall test the interface circuitry and output logic.

3.4.2.6 Control, Logic-Timing C-9506( )/AWG-9. Operation of the logic and timing control (LTC) shall be verified by performing the following tests.

3.4.2.6.1 BIT verification. The CTS shall test the built-in-test circuitry of the LTC.

3.4.2.6.2 Missile identification. The CTS shall test the missile identification function.

3.4.2.6.3 Missile ready preparation. The CTS shall test the missile ready preparation function. This shall include test of the timing of the events leading up to application and maintenance of power command signals to the AIM-7 and AIM-54A missiles.

3.4.2.6.4 Tuning. The CTS shall have the capability for performing a test of the tuning logic in the LTC. This sequence of tests shall occur during the timing period of the missile ready preparation.

3.4.2.6.5 Missile ready functions. The CTS shall test the missile ready functions after having completed missile ready preparation and AIM-7 Missile tuning. The signals appropriate to the missile ready functions being simulated shall be applied and maintained by the CTS to verify the LTC function. Tests shall be performed for verification of the injected video signals.

3.4.2.6.6 Missile auxiliaries subsystem (MAS)/coded integrated armament control system (CIACS) interface logic. The CTS shall test the MAS/CIACS interface. This LTC capability shall be verified by receiving and sending digital data via the station control channel and by observing the inputs and outputs of this interface.

3.4.2.6.7 Launch initiation. The CTS shall test the launch initiate output signal from the LTC when all launch prerequisite conditions have been satisfied.

3.4.2.6.8 Launch cycle. The CTS shall test for the sequence of signals provided by the LTC during the launch cycle for the AIM-54A Missile and for the AIM-7 missiles.

3.4.2.6.9 MOAT. The CTS shall test the MOAT logic in the LTC. All timing functions of MOAT shall be tested and the sequencing of these functions shall be under the control of the CTS. The CTS shall monitor the test report lines and determine pass or fail conditions.

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3.4.2.6.10 Training. The CTS shall test LTC input and output signals when in the training select mode.

3.4.2.6.11 Target doppler reference frequency (TDRF). The CTS shall have the capability of testing the TDRF function of the AIM-54A Missile.

3.4.2.6.12 Dc analog functions. The CTS shall have the capability of testing the dc analog functions associated with the AIM-54A and AIM-7 Missiles.

3.5 Workmanship. Applicable.

3.6 Elements.

3.6.1 Console, Test, Computer OJ-256( )/AWM-23(V). The computer test console (bay no. 1) shall provide the test capabilities for testing the missile auxiliaries WRAS and the PS/S.

3.6.1.1 Dummy Load, Electrical DA-633( )/AWM-23(V). The electrical dummy load shall provide resistive loads for testing the PS/S.

3.6.1.2 Panel, Test, Electrical C-9180( )/AWM-23(V). The electrical test panel shall have the switches, controls, and monitoring capabilities required for test of the PS/S. Indicators shall be provided to allow constant monitoring of the prime power and the bulk store status of the PS/S. Switches shall be provided to facilitate testing the bulk store functions and troubleshooting the power supply functions.

3.6,1.3 Mounting MT-4504( )/AWM-23(V). The WRA mounting shall be a dual mounting used in three test configurations. When testing an LTC the relay matrix loop test adapter shall be stored in bay no. 4 and the mounting shall hold the LTC and the missile auxiliaries simulator and shall provide the interfaces between the CTS and these units. When testing a signal data converter, the missile auxiliaries simulator and the relay matrix loop test adapter shall be stored in the WRA stowage area provided in bay no. 4 and the signal data converter shall be installed in its place on the mounting. The mounting shall provide the interface between the CTS and the signal data converter under test. When neither the signal data converter nor LTC is being tested, the mounting shall hold the missile auxiliaries simulator and the relay matrix loop test adapter for self test.

3.6.1.4 Simulator, missile auxiliaries. The missile auxiliaries simulator shall be used when testing an LTC (3.6.1.3). The simulator shall provide all of the interfaces normally seen by an LTC when operating in the missile auxiliaries subsystem.

3.6.1.5 Power Supply Assembly PP-6851( )/AWM-23(V). The power supply assembly shall contain modular type power supplies to provide the power forms required for the signal data converter and LTC, and also the missile auxiliaries test logic.

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3.6.1.6 Simulator, Electronic Command Signal SM-657( )/AWM-23(V). The electronic command signal simulator shall contain the logic necessary to control and monitor either the signal data converter or LTC WRAs. The drawer shall also contain the simulation functions needed for simulating either the signal data converter WRA or the LTC WRA.

3.6.1.7 Housing connector assembly - loop test, relay matrix. The housing connector assembly shall provide circuitry to permit self-test of the relay switched signal channels when an LTC WRA is not connected. A dc voltage shall be applied to all signal channels through a buffering resistor to allow each channel to be individually switched and monitored.

3.6.2 Console, Test, Computer OJ-257( )/AWM-23(V). Computer test console (bay no. 2) shall operate in conjunction with computer test console (bay no. 3) to provide test positions for testing of the computer subsystem WRAs.

3.6.2.1 Ac and dc radiometer - voltmeter (DRVM). A DRVM, Dana, 5743-S-2181A, or equivalent, shall be supplied and shall be capable of measuring resistance, ac and dc voltages and ac voltage ratios to test the WRAs under automatic station control.

3.6.2.2 Programmable counter/timer. A programmable counter/timer, Monsanto 110B-021B, or equivalent, shall be supplied and shall be capable of measuring frequency, pulse width, pulse count, and time intervals to test the WRAs under automatic station control.

3.6.2.3 Indicator, Data Display IP-1129( )/AWM-23(V). The data display indicator shall be capable of presenting visual displays of test station functions, status, and test data which are to be monitored by the operator.

3.6.2.4 Mounting MT-4505( )/AWM-23(V). The WRA mounting shall be a dual carrier used in three WRA configurations. When testing a data processor the mounting shall hold the data processor and the power supply and bulk store loop test adapter (3.6.2.6) and shall provide the interface between the CTS and these units. When testing a PS/S, the WRA mounting shall hold the data processor simulator and the PS/S and shall provide the interface between the CTS and these units. When neither WRA is being tested, the mounting shall hold the power supply and bulk store loop test adapter and the data processor simulator for self-test.

3.6.2.5 Simulator, data processor. The data processor simulator shall provide an interface with the digital computer for testing the input/output channel of the digital computer, with the PS/S for testing the bulk store functions and with the system interface simulator (SIS) for self-test of the SIS. When a tactical data processor is to be tested, the data processor simulator shall be stored in the WRA stowage area provided in bay no. 4.

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3.6.2.6 Adapter, power supply and bulk store - loop test. The power supply and bulk store loop test adapter shall provide the circuitry to test the CT'S interfaces utilized to test a power supply storer. When a data processor is to be tested, the adapter shall be located on Mounting MT-4505( )/AWM-23(V). When a PS/S is to be tested, the adapter shall be stored in the WRA stowage area provided in bay no. 4.

3.6.2.7 Switching Unit, Fault Isolation SA-1898( )/AWM-23(V). The fault isolation switching unit shall contain relays that switch signals to the counter-timer or DRVM under computer control.

3.6.2.8 Simulator, Electronic Command Signal Generator SM-659( )/AWM-23(V). The electronic command signal generator simulator shall contain the circuits and switching networks required for simulation of the Weapon Control System (WCS) and avionics/data processor interface.

3.6.3 Console, Test, Computer 0J-258( )/AWM-23(V). The computer test console (bay no. 3) shall operate in conjunction with bay no. 2 to provide test positions for testing of the computer subsystem WRAs. The bay shall also provide the control logic for controlling the functional operation of the station.

3.6.3.1 Oscilloscope. An oscilloscope, Hewlett-Packard 180AR/1801A-90/1821A, or equivalent, shall be supplied and shall provide the capabilities required for waveform analysis when manually testing the WRAs and for performing station maintenance tests.

3.6.3.2 Panel, Test, Electrical C-9181( )/AWM-23(V). The electrical test panel shall provide control functions for testing the digital computer and shall have a keyboard for manual entry of data into-the computer. The panel shall provide indications of status of the tape reader and the computer.

3.6.3.3 Mounting MT-4506( )/AWM-23(V). The mounting shall hold the digital computer WRA and shall provide the interfaces between the CTS and this WRA. A slave digital computer shall be in this mounting when a tactical digital computer is not under test. When a tactical digital computer is being tested, the slave digital computer shall be stored in the WRA storage area provided in bay no. 4.

3.6.3.3.1 Computer, Digital CP-1159( )/AWG-9. The slave digital computer shall perform the functions of a central processor for the test station. Normally this slave digital computer will be mounted in the WRA mounting. When a tactical digital computer is to be tested, this slave digital computer will be stowed in the storage location of bay no. 4.

3.6.3.4 Mounting MT-4507( )/AWM-23(V). The WRA mounting shall be capable of holding either the memory unit or the memory unit simulator and shall provide the interfaces between the CTS and either device.

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3.6.3.5 Simulator-memory unit. The memory unit simulator shall provide a simulated memory unit interface with the digital computer. The memory unit simulator shall be stored in the WRA stowage area provided in bay no. 4 when a memory unit is installed for test.

3.6.3.6 Signal driver memory unit. The signal driver memory unit shall contain the circuit necessary to load the memory unit memory. This assembly is connected to the memory unit simulator or the memory unit under test.

3.6.3.7 Power Supply Assemblies PP-6852( )/AWM-23(V), PP-6853( )/AWM-23(V), and PP-6854( )/AWM-23(V). The power supply assemblies shall contain modular type power supplies to provide the power forms required for the digital computer, memory unit, and data processor for testing.

3.6.4 Console, Test, Computer OJ-259( )/AWM-23(V). The computer test console (bay no. 4) shall provide primary power control, station magnetic tape reader, storage, and secondary power to the CTS.

3.6.4.1 Panel, Power Distribution SB-3697( )/AWM-23(V). The power distribution panel shall control and monitor station power, and shall indicate power or cooling malfunctions.

3.6.4.2 Magnetic tape reader. A magnetic tape reader, Peripheral Equipment Corp. 3510-72, or equivalent, shall be supplied to load the test, tactical and BIT programs which are stored on magnetic tape (see 3.2.12).

3.6.4.3 Control, logic-magnetic tape reader C-9182( )/AWM-23(V). The magnetic tape reader logic control shall contain the control and data formatting logic for operating the magnetic tape reader.

3.6.4.4 Work surface assembly. The work surface assembly shall provide a work station for maintenance.

3.6.4.5 Drawer, storage. A storage space shall be provided below the work surface and shall have a minimum storage volume of 0.75 cubic feet.

3.6.4.6 Power Supply Assemblies PP-6855( )/AWM-23(V), PP-6856( )/AWM-23(V) and PP-6857( )/AWM-23(V). The power supply assemblies shall supply the major secondary power forms required by CTS logic circuits.

3.6.5 Elapsed time meter. An elapsed time meter, conforming to MIL-M-7793, shall be provided. The meter shall be readily visible during operation of the CTS and shall be capable of registering at least 9,999 hours.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Applicable.

4.2 General. Applicable.

4.2. 1 Classification. Applicable.

4.2.2 Individual items. Applicable. The Computer Test Station shall include program tapes for test station self-test and functional and diagnostic programs for applicable AN/AWG-9 WRAs.

4.2.3 In-process inspection. Applicable.

4.2.4 In-process data. Applicable.

4.2.5 Acceptance test plan. Applicable. Table IV shall apply.

4.2.6 Test results. Applicable.

4.2.7 Acceptance. Applicable.

4.2.8 Utilities. Applicable.

4.2.9 Configuration. Applicable.

4.3 Tests.

4.3.1 Electrical and electronic items - operational run-in test. Applicable.

4.5.2 Mechanical items. Applicable.

4.4 Environmental test conditions. Applicable.

4.5 Test location. Applicable.

4.6 Acceptance schedule. Applicable.

4.7 Test procedure. Applicable.

4.8 Rejection and retest. Applicable.

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TABLE IV

ACCEPTANCE TEST PLAN FOR COMPUTER TEST STATION

OJ-216( )/AWM-23(V)

1.0 TEST OBJECTIVE. To demonstrate self-test and operational run-in tests of the Computer Test Station.

2.0 TEST CONDITIONS

a. AWG-9 Slave Units

(1) 481451-1XX

b. Utilities Required (Information only, ref 4.2.8)

(1} Power as specified in 3.2.2.1
(2) Clean air cooling as specified in 3.2.2.2

3.0 TESTS. In accordance with Acceptance Test Procedure 3282207,

a. Perform self-test.

b. Perform operational run-in tests.

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5. PREPARATION FOR DELIVERY

5.1 General. Applicable.

6. NOTES

6.1 Intended use. The CTS is intended for intermediate level use to functionally test, fault isolate, load memory and verify the serviceability of certain WRAs of the computer and missile auxiliaries subsystems of Weapon Control System (WCS) AN/AWG-9.

6.2 Ordering data. Applicable.

6.3 Definitions. Applicable.

6.4 Nomenclature cross reference. The following is a cross reference between the official Navy nomenclature and the major contractor part numbers. The contractor part number suffix is controlled by the configuration list in the contract.

	Official Nomenclature	HAC Part No.
a.	Computer, Digital CP-1159( )/AWG-9	481451-1XX
b.	Memory Unit MU-588( )/AWG-9	481452-1XX
c.	Processor, Data CV-3151( )/AWG-9 and CV-3165( )/AWG-9	481461-1XX
d.	Power Supply-Storer PP-6776( )/AWG-9	481462-1XX
e.	Converter, Signal Data CV-3152( )/AWG-9	481710-1XX
f.	Control, Logic-Timing C-9506( )/AWG-9	481720-1XX
g.	Test Station, Computer OJ-216( )/AWM-23(V)	493400-1XX
h.	Console, Test, Computer OJ-256( )/AWM-23(V)	493421-1XX
i.	Console, Test, Computer OJ-257( )/AWM-23(V)	493422-1XX
j.	Console, Test, Computer OJ-258( )/AWM-23(V)	493423-1XX
k.	Console, Test, Computer OJ-259( )/AWM-23(V)	493424-1XX
l.	Kit, Tool, Special TK-222( )/AWM-23(V)	493601-1XX

6.5 Liquid cooler. Not applicable.

6.6 Use of this specification. This specification is a detail specification which uses MIL-T-81734(AS) as a general specification. Where the corresponding paragraphs of MIL-T-81782(AS) are the same as MIL-T-81734 (AS) the word "applicable" is used in this specification. When the term "not applicable" is used in this specification it means that the corresponding paragraph requirements of MIL-T-81734(AS) are not required to be met (see 6.6 of MIL-T-81734(AS)).

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6.7 Abbreviations and acronyms. Applicable. In addition, the following shall apply:

ADA	Analog-digital-analog
CIACS	Coded integrated armament control system
CTS	Computer test station
D/A	Digital-to-analog
DRO	Destructive readout
DRVM	Digital radiometer voltmeter
HP RF	High pulse repetition frequency
IFF	Identification, friend or foe
IR	Infrared
LTC	Logic and timing control
MAS	Missile auxiliaries subsystem
MOAT	Missile on aircraft test
MTM	Magnetic tape memory
NDRO	Nondestructive readout
PS/S	Power supply/storer
SIS	System interface simulator
SSC	Start sample command
SSCD	Start sample command delayed
SSI	Standard serial interface
TDRF	Target doppler reference frequency

6.8 Convenience outlet power. Applicable.

6.9 Changes from previous issue are not marked.

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DOCUMENT IDENTIFIER AND TITLE
MIL-T-81782C(AS) Test Station, Computer OJ-216( )/AWM-23(V)

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A GIVE PARAGRAPH NUMBER AND WORDING

   B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

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